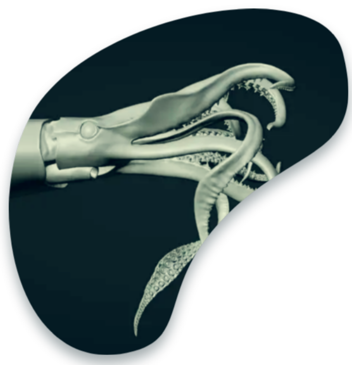

Risk Analysis

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Category: Working paper summary

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1 Documents

Two complementary working papers submitted to the 11th Meeting of the IATTC Scientific Advisory Committee (2020):

- **SAC-11 INF-F** (Maunder et al. 2020) — the *methods* paper: *Implementing reference point-based fishery harvest control rules within a probabilistic framework that considers multiple hypotheses.*
- **SAC-11-08** (Aires-da-Silva et al. 2020) — the *application* paper: *Risk analysis for management of the tropical tuna fishery in the Eastern Pacific Ocean, 2020.*

2 How the two papers relate

The two documents are complementary halves of a single body of work. **SAC-11 INF-F** (Maunder et al. 2020) sets out the *general methodology*: how to implement a reference-point-based harvest control rule (HCR) inside a probabilistic framework that carries multiple competing stock-assessment models rather than a single "best" model, using eastern Pacific Ocean (EPO) bigeye tuna as a worked case study. **SAC-11-08** (Aires-da-Silva et al. 2020) *applies* that methodology to both EPO bigeye and yellowfin tuna and reports the resulting stock-status and management-decision probabilities.

Both papers explicitly frame the approach as a **pragmatic compromise between computational demand, complexity, and statistical rigour**. They weight models using information contained in the *data*, the *population processes*, and the *fishery processes*, supplemented by *expert judgement*, and both openly acknowledge that the complexity of contemporary integrated stock-assessment models prevents strict adherence to ideal statistical criteria (e.g. likelihood- or AIC-based model selection, full Bayesian MCMC). The weighting scheme is therefore deliberately semi-quantitative and partly subjective by design.

The motivation was concrete: by 2018–2019 the IATTC staff had concluded that its "best assessment" (single base-case) results for bigeye and then yellowfin were **too sensitive to new data** — particularly recent longline abundance-index observations — to support management advice. Rather than pick one model, the staff moved to a **risk analysis** that carries a suite of plausible alternative models ("states of nature"), weights them by plausibility, and produces probability statements about whether the HCR's target and limit reference points have been exceeded.

3 Conceptual foundation common to both papers

The management problem. The IATTC HCR for tropical tunas (Resolution C-16-02) is written in probabilistic terms: if the probability that fishing mortality F exceeds the limit reference point F_{LIMIT} is greater than 10%, management measures must be established that give at least a 50% probability of reducing F to the target F_{MSY} or below, and less than a 10% probability that F exceeds F_{LIMIT}. Analogous conditions apply to spawning biomass S relative to S_{MSY} and S_{LIMIT}. Evaluating such a rule *requires* an explicit quantification of uncertainty — a point estimate alone cannot answer it.

Definition of risk. Following FAO (1995) (FAO 1995), risk is defined as "the probability of something undesirable happening" — here, the probability of exceeding the target or limit reference points.

Two categories of uncertainty addressed. Both papers concentrate on (1) **parameter uncertainty** — the estimation precision of quantities within a given model — and (2) **model-structure uncertainty** — the possibility that different model structures each give a reasonable representation of reality. A third category, **uncertainty about the future** (process variation), is explicitly deferred to later development and to Management Strategy Evaluation (MSE).

Quantities of interest are ratios. Because status is judged relative to reference points, the probability distributions are computed for the *ratios* (e.g. F_{cur}/F_{MSY}, S_{cur}/S_{MSY}), not for the individual components.

Idealized vs. practical approach. SAC-11 INF-F first describes an *idealized* fully-Bayesian solution — integrate all data and data-based priors into models, parameterize structural uncertainty, run MCMC, and obtain posterior distributions for the quantities of interest — and then explains why it is not currently feasible for tropical tunas: AIC and likelihood-based selection behave poorly for complex, highly parameterized models whose assumptions are routinely violated (Burnham and Anderson 2002); not all structural hypotheses can be reduced to a set of estimable parameters; data are often insufficient to estimate the relevant parameters; and MCMC over many models is prohibitively slow. The **practical approach** is the paper's answer to these obstacles.

4 The methodology in SAC-11 INF-F (the "practical approach")

The practical approach has **five main steps**: (1) establish a hierarchy of hypotheses and models; (2) define a weighting system for hypotheses and models; (3) calculate probability distributions of the quantities of interest for each model; (4) combine those distributions across models; and (5) present the results as a risk analysis.

4.1 Step 1 — A three-level hierarchy of hypotheses

Hypotheses about the state of nature are represented by stock-assessment models, arranged in a **flow chart / hierarchical tree** with three levels. The hierarchy exists to prevent any single hypothesis — or an "overarching" hypothesis represented by many models — from inadvertently dominating the result, and to make model development and weight assignment orderly.

- **Level 1 — Overarching hypotheses.** Broad states of nature (e.g. for bigeye: *the recruitment regime shift is real* vs. *the regime shift is an artefact of model misspecification*). These are hard or inappropriate to distinguish by fit to data, so they are weighted **by expert opinion only**, and all lower-level weights are conditional on them.
- **Level 2 — Hypotheses.** Specific hypotheses that *can* be represented by a model and differentiated by fitting to data. Level 2 is divided into sub-levels (A, B, …), each addressing a distinct assessment issue (for bigeye, 2A = the recruitment regime shift; 2B = misfit to length-composition data for the fishery with asymptotic selectivity). Models often must be combined across sub-levels to resolve all issues at once.
- **Level 3 — Sub-hypotheses.** Evaluated differently from Level 2, to avoid the influence of data or to reduce the number of analyses. The canonical case is **steepness (h)** of the Beverton–Holt stock–recruitment relationship: because steepness is notoriously biased when estimated inside assessment models, it is fixed at a discrete set of values (h = 0.7, 0.8, 0.9, 1.0) and weighted only by expert opinion and convergence, so the data are not allowed to inform it.

4.2 Step 2 — The weighting system

Weights convert the hierarchy into usable probabilities. The system has four parts: (a) discrete weight categories; (b) a set of weight metrics; (c) assignment and rescaling of weights; and (d) reduction of the number of models.

(a) Discrete weight categories. To tame subjectivity, weights are drawn from four discrete categories rather than a continuous scale — **None = 0, Low = 0.25, Medium = 0.5, High = 1.0** — while acknowledging that the numerical values themselves are subjective choices.

(b) Weight metrics. Each model's overall weight is the **product** of several component metrics (Equation 1):

$$W(\text{model}) = W(\text{Expert}) \times W(\text{Convergence}) \times W(\text{Fit}) \times W(\text{Plausible parameters}) \times W(\text{Plausible results}) \times W(\text{Diag}$$

- **W(Expert)** — subjective expert judgement assigned *a priori*, drawing on experience with this and related assessments and on population-dynamics theory.
- **W(Convergence)** — whether the estimation converged; a non-positive-definite Hessian is treated as non-convergence and given **zero weight**, which eliminates the model.
- **W(Fit)** — overall fit to data, based on **AIC** but with deliberately *liberal* criteria so that all models are retained rather than eliminated; the best-fitting (lowest-AIC) model gets *High* and the worst gets *Low*, with a linear scale between. Special rules handle models that use different data, time frames, or data-weighting so AIC is not compared unfairly (e.g. computing AIC only over data common to all models).
- **W(Plausible parameters)** — realism of the estimated additional parameters (e.g. an implausible growth or M value flags misspecification being absorbed by that parameter).
- **W(Plausible results)** — realism of derived results (e.g. an unrealistically high or low estimated fishing mortality). Flagged as highly subjective; care is needed that it not control the outcome.
- **W(Diagnostics)** — model reliability from a suite of diagnostics. Uniquely, this component is a **sum** of sub-weights (Equation 2) so that no single diagnostic is over-weighted:

$$W(\text{Diagnostics}) = W(\text{ASPM}, R_0, \text{Catch curve}) + W(\text{Retrospective}) + W(\text{Composition residuals}) + W\text{(Index resid}$$

The diagnostic sub-components probe specific failure modes: the R_0 **likelihood profile** and **ASPM / ASPM-Rdev** and **catch-curve** diagnostics test whether abundance information comes from the indices (preferred) or is being driven by composition data or misspecification; **retrospective analysis** tests whether estimates drift systematically as data years are added; and residual diagnostics for composition, index and recruitment test violations of likelihood assumptions.

(c) Assignment and rescaling. Weights are assigned following the hierarchy and then **rescaled to sum to one** so they behave as probabilities. Crucially, rescaling is done **conditionally along the branches of the tree** — Level 1 across all overarching hypotheses, Level 2 within each sub-level branch, Level 3 within each Level-2 branch — turning the flow chart into a **probability tree**. This prevents an overarching hypothesis from gaining weight merely because more models happen to represent it. Branch-specific metrics (e.g. W(Expert)) are assigned within a branch; metrics with a common interpretation across the tree (e.g. W(Plausible results)) are assigned globally. Where consensus among experts is absent, each expert's weights are rescaled and then averaged so each expert has comparable influence.

(d) Reducing the number of models. Because diagnostics like the R_0 profile and retrospective analysis require multiple model runs, the number of models must be kept practical. Any metric scored *None* (zero) zeroes the product and eliminates the model. Defining a simpler **"Base" model** for each Level-2A branch lets whole groups of dependent models be eliminated without running them all — if a Base model fails (e.g. non-convergence), its derived Level-2B/Level-3 models are dropped too. Checking W(Expert) and W(Convergence) *first* saves computing the remaining metrics for models that will be eliminated anyway.

4.3 Step 3 — Probability distribution of the quantities of interest, per model

A fully Bayesian per-model posterior would be preferred but is impractical at scale, so **two approximations** are used: (i) frequentist approximations of the posterior for the quantity given the model (justified because with informative data good frequentist coverage approximates a Bayesian posterior), and (ii) **normal approximations** based on the estimated standard error of the (often derived) quantity, since a profile likelihood on a derived ratio may not be obtainable. The per-model distribution is then rescaled to a proper density $P(\text{Quantity} \mid \text{Model} = m)$.

4.4 Step 4 — Combining distributions across models (model averaging)

The distributions are combined by **model averaging** (Hoeting et al. 1999) weighted by the rescaled model probabilities (Equation 6):

$$P(\text{Quantity}) = \sum_{m \in \text{Models}} P(\text{Quantity} \mid \text{Model} = m)\, P(\text{Model} = m)$$

evaluated over a grid of the quantity fine and wide enough to resolve the tails (which matter because the HCR turns on tail probabilities such as $P(F > F_{LIMIT}) > 0.1$). The **cumulative distribution function** is then formed by trapezoidal integration (Equation 7) and used to read off the probability of exceeding a reference point; where the normal approximation is used, the normal CDF is used directly.

4.5 Step 5 — Presenting the risk analysis

Results are presented as: probability density plots of the ratios (optionally decomposed by weighting component to show each factor's influence); CDFs for reading exceedance probabilities; **Kobe (phase) plots** placing each model and the combined result relative to reference points; and **decision tables** (in the Punt & Hilborn 1997 format) giving the performance measure — the exceedance probability — for each management action under each state of nature, with a weighted-average column. **Sensitivity analysis to the weights** is recommended to show how robust the advice is to the (subjective) weighting.

4.6 The bigeye case study and its lessons

Applying this to EPO bigeye, the hierarchy was built around two issues: the mid-1990s **recruitment regime shift** (Level 1: real vs. artefact) and **misfit to longline length-composition data** under assumed asymptotic selectivity. Two application-specific metrics were added — **W(Fix regime)** (a model's ability to remove the regime shift, measured by the ratio of late- to early-period median recruitment) and **W(Empirical selectivity)** (agreement between model-implied and "empirical" selectivity). Several base models were eliminated (e.g. the Ricker model for non-convergence). The discussion documents the framework's honest limitations: pervasive **subjectivity** in hypothesis choice and weighting; the tension of using standard statistical machinery for *parameter* uncertainty while deliberately setting it aside for *model-structure* uncertainty; poorly estimated **tails** under the normal approximation (MCMC comparisons showed fatter, sometimes asymmetric real tails); and, most consequentially, **bimodality** in the combined distributions — the bigeye result sits between an optimistic and a pessimistic cluster of models that available data cannot distinguish.

5 The methodology in SAC-11-08 (the application)

SAC-11-08 (Aires-da-Silva et al. 2020) condenses the INF-F methodology into a **four-step "pragmatic risk analysis"** and applies it to both bigeye and yellowfin:

1. **Identify alternative hypotheses ("states of nature")** about population dynamics that address the main assessment issues, arranged in a flow chart showing dependencies (details for yellowfin in SAC-11 INF-J; for bigeye in SAC-11 INF-F).
2. **Translate hypotheses into stock-assessment models** — 12 model configurations for yellowfin and 14 for bigeye, each run at four steepness values (h).
3. **Determine the relative weight of each model** as a relative probability, using the mix of metrics from INF-F (expert opinion, model fit, plausibility of parameters and results, diagnostics, convergence), then rescale to relative probabilities — explicitly to avoid the bias of equal weighting.
4. **Combine model relative probabilities with the per-model probability distributions** of the quantities of interest to compute the probability that the C-16-02 target and limit reference points are exceeded.

The analysis is run **separately for each species** and split into two components: **current stock status** (weighted-average ratios and exceedance probabilities via combined CDFs) and **decision analysis** (the consequences of six alternative purse-seine closure durations — 0, 36, 70, 72, 88, 100 days — on the F-based reference points, with F assumed proportional to open fishing days adjusted for the October "corralito" spatial closure and fleet-capacity changes). No projections were run, so the spawning-biomass reference points could not be evaluated in the decision analysis — a deferred development.

5.1 Headline results

- **Yellowfin** — 48 models, all retained. The combined distributions are **unimodal**. $P(F_{cur} > F_{MSY}) \approx 9\%$, $P(S_{cur} < S_{MSY}) \approx 12\%$, and P of breaching either **limit** reference point $\approx 0\%$. The stock is unambiguously healthy; the current 72-day closure carries only a 9% risk of exceeding F_{MSY}.

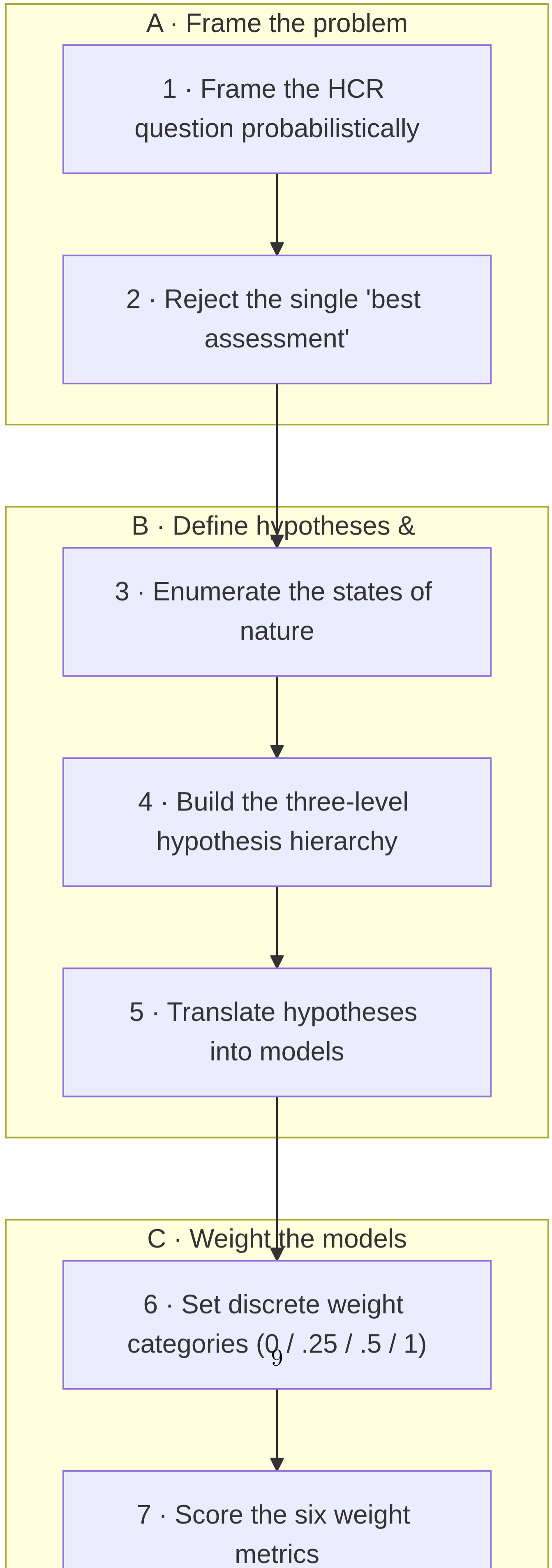
- **Bigeye** — 44 of 48 models retained (4 failed to converge). The combined distributions are **bimodal**: $P(F_{cur} > F_{MSY}) \approx 50\%$, $P(S_{cur} < S_{MSY}) \approx 53\%$, with non-negligible limit-exceedance probabilities ($\approx 5\%$ for F, $\approx 6\%$ for S). The optimal management direction cannot be determined from the analysis alone. However, the combined *pessimistic* models give only a ~10% probability of exceeding F_{LIMIT} at the current 72-day closure, so a **status-quo harvest strategy is judged appropriate in the short term**.

The paper cautions that, for the bigeye bimodal case, simple summaries such as the expected value or a single $P(F_{cur} > F_{MSY})$ statement should be **avoided** in favour of the whole distribution, and neither the average nor a single assumed state of nature should drive the closure decision without weighing the risk of that assumption being wrong. It recommends continued improvement of the assessments (growth estimation — which carried ~58% of the model weight — model time-span, stock structure, natural mortality, selectivity) and development of **MSE** (Punt et al. 2016), for which the weighted model set can seed the operating models.

6 Key-points workflow (the procedure at a glance)

The following captures the end-to-end workflow common to both papers, from problem framing to management advice.

1. **Frame the management question probabilistically.** Start from the HCR (C-16-02), which is written as probability statements about F and S relative to target (F_{MSY}, S_{MSY}) and limit (F_{LIMIT}, S_{LIMIT}) reference points; the analysis must deliver those probabilities.
2. **Reject the single "best assessment."** Because point estimates cannot express the uncertainty the HCR needs — and because the base-case results were over-sensitive to new data — adopt a multi-model risk analysis instead.
3. **Enumerate the states of nature.** Identify the main unresolved assessment issues (bigeye: recruitment regime shift, composition misfit; yellowfin: index inconsistencies, size increases, composition misfit, possible spatial structure) and cast them as competing hypotheses.
4. **Build the three-level hypothesis hierarchy.** Level 1 overarching hypotheses (expert-weighted only) → Level 2 data-distinguishable hypotheses in issue-specific sub-levels → Level 3 sub-hypotheses (e.g. fixed steepness values). Draw it as a flow chart / probability tree.
5. **Translate hypotheses into models.** Configure a stock-assessment model for each hypothesis (12 configs × 4 h for yellowfin; 14 configs × 4 h for bigeye), combining sub-level models where an issue needs more than one change.
6. **Set up the weighting system.** Use discrete weight categories (None 0 / Low 0.25 / Medium 0.5 / High 1.0) to limit subjectivity.


A · Frame the problem
1 · Frame the HCR question probabilistically
2 · Reject the single 'best assessment'
B · Define hypotheses &
3 · Enumerate the states of nature
4 · Build the three-level hypothesis hierarchy
5 · Translate hypotheses into models
C · Weight the models
6 · Set discrete weight categories (0 / .25 / .5 / 1)
7 · Score the six weight metrics

7. **Score each weight metric.** Assign W(Expert), W(Convergence), W(Fit) (liberal AIC-based), W(Plausible parameters), W(Plausible results), and W(Diagnostics) (itself the *sum* of R_0/ASPM/catch-curve, retrospective, and residual sub-weights). Add application-specific metrics as needed (bigeye: W(Fix regime), W(Empirical selectivity)).
8. **Eliminate implausible models efficiently.** Any *None* score zeroes the product; check Expert and Convergence first; use "Base" models so failure of a base eliminates its dependent branch without running every model.
9. **Compute each model's overall weight.** Take the product of the (summed, for diagnostics) component weights — Equation 1.
10. **Rescale weights conditionally along the tree.** Rescale to sum to one within each branch and level so the flow chart becomes a probability tree; this stops any hypothesis from being over-weighted simply for being represented by more models.
11. **Build each model's probability distribution for the quantity of interest.** Work with ratios (e.g. F_{cur}/F_{MSY}); approximate the per-model posterior with frequentist / normal approximations based on the estimated standard error, then rescale to a density.
12. **Model-average across models.** Combine per-model densities weighted by rescaled model probabilities — Equation 6 — over a grid fine and wide enough to capture the tails.
13. **Form CDFs and read exceedance probabilities.** Integrate (trapezoidal rule, Equation 7) to get P(exceed reference point) currently and under each management action.
14. **Run the decision analysis.** Evaluate alternative purse-seine closure durations (0/36/70/72/88/100 days), translating closure days into F (proportional to open days, adjusted for the corralito and fleet capacity), and tabulate exceedance probabilities per model and combined.
15. **Present the results.** Density plots (optionally decomposed by weighting factor), CDFs, Kobe phase plots (with the combined point defined by $P = 0.5$), and Punt–Hilborn decision tables; run **sensitivity analysis on the weights**.
16. **Interpret with the uncertainty made explicit.** Report probability statements, not point estimates; where the combined distribution is bimodal (bigeye), present the whole distribution and the optimistic/pessimistic states rather than a misleading average.
17. **Feed forward to improvement and MSE.** Use the hypotheses that drive bimodality and those carrying high weight to prioritise assessment research (growth, time-span, M, selectivity, stock structure), and reuse the weighted model set as operating models for Management Strategy Evaluation.

7 Points of emphasis and acknowledged limitations

- **A pragmatic, not idealized, method.** Both papers position the approach explicitly as a compromise among computational demand, complexity, and statistical rigour — a deliberate step back from full Bayesian/MCMC and from likelihood- or AIC-only model selection, which behave poorly for complex, highly parameterized, assumption-violating assessment models.

- **Model fit plays only a limited role.** Fit (AIC) is *one* metric among many and is applied with liberal criteria; reliability, plausibility, diagnostics, and expert judgement carry substantial weight. This is a defining feature, not an oversight.
- **Subjectivity is explicit and bounded, not eliminated.** Hypothesis selection, category values, and several metrics are subjective; discrete categories and conditional rescaling are the tools used to contain that subjectivity, and weight sensitivity analysis is used to test it.
- **Parameter vs. structural uncertainty are handled inconsistently by necessity.** Standard statistical machinery is used for parameter uncertainty but deliberately set aside for model-structure uncertainty — an acknowledged internal tension.
- **Tails and bimodality are the weak points.** Normal approximations may misrepresent asymmetric tails (which the HCR depends on), and the bigeye bimodality — two data-indistinguishable states of nature — limits how far the risk analysis can guide management. Both are flagged for future work via improved assessments and MSE.

8 References

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